

Embedding Measures for ℓ_A^p When $p > 2$

A full candidate solution to the measure question in arXiv:2210.05823

Automated mathematics run `fa_banach_001`, lane 6

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Status. Full solution candidate, likely valid, awaiting human review. The result fully characterizes the norm inequality asked about at the end of Cheng–Felder for every finite positive Borel measure, and hence for the particular atomic measure in their equation (7.2).

1 Source problem

The source is R. Cheng and C. Felder, *On Interpolation by Functions in ℓ_A^p* , arXiv:2210.05823 [1]. For $1 < p < \infty$, the space ℓ_A^p consists of analytic functions

$$f(z) = \sum_{n=0}^{\infty} a_n z^n, \quad \|f\|_{\ell_A^p} = \left(\sum_{n=0}^{\infty} |a_n|^p \right)^{1/p}.$$

The source asks for a full characterization, when $2 < p < \infty$, of its equation (7.2), reproduced from page 27 below.

The closing question occurs on page 35.

In measure language, the problem is to characterize those finite positive Borel measures μ on $\mathbb{D}$ for which

$$\int_{\mathbb{D}} |f(z)|^p d\mu(z) \leq C \|f\|_{\ell_A^p}^p \quad (f \in \ell_A^p). \quad (1)$$

Indeed, equation (7.2) is (1) for

$$\mu_Z = \sum_{k \geq 0} (1 - |z_k|^q)^{p-1} \delta_{z_k}, \quad \frac{1}{p} + \frac{1}{q} = 1. \quad (2)$$

For $0 < h \leq 1$ and $\theta \in \mathbb{R}$, write

$$S(\theta, h) = \{re^{it} \in \mathbb{D} : 1 - h \leq r < 1, |t - \theta| \leq h/2\},$$

with angles understood modulo 2π . Define

$$[\mu]_{p-1} = \sup_{\theta \in \mathbb{R}, 0 < h \leq 1} \frac{\mu(S(\theta, h))}{h^{p-1}}.$$

2 Proof intuition

The correct exponent can be read from a localized Dirichlet kernel. A polynomial with N equal coefficients, normalized in ℓ^p , has size $N^{1-1/p}$ throughout a Carleson square of side comparable to N^{-1} . If (1) holds, that square can therefore carry mass at most $N^{-(p-1)}$.

THEOREM 7.1. Assume $1 < p < \infty$ and $1/p + 1/q = 1$, and $\{z_0, z_1, z_2 \dots\}$ is a sequence of distinct points in $\mathbb{D}$. There exists $K > 0$ such that

$$(7.2) \quad \sum_{k=0}^{\infty} (1 - |z_k|^q)^{p-1} |f(z_k)|^p \leq K \|f\|_p^p$$

for all $f \in \ell_A^p$, if and only if (UB) holds, that is, there exists $C > 0$ such that

$$(7.3) \quad \left\| \sum_{k=0}^{\infty} c_k \Lambda_{z_k} \right\|_q \leq C \left(\sum_{k=0}^{\infty} |c_k|^q \|\Lambda_{z_k}\|_q^q \right)^{1/q}$$

for all constants $\{c_k\}_{k=0}^{\infty}$.

PROOF. If (7.2) holds, then for any constants $\{c_k\}_{k=0}^{\infty}$,

$$\left\| \sum_{k=0}^{\infty} c_k \Lambda_{z_k} \right\|_q = \sup_{f \neq 0} \frac{\left| \sum_{k=0}^{\infty} c_k \Lambda_{z_k}(f) \right|}{\|f\|_p}$$

Figure 1: Equation (7.2) and its interpretation in the source, page 27.

It remains, therefore, to fully characterize the norm inequality (7.2) when $2 < p < \infty$. Looking back to earlier sections, we would like to uncover any connection between weak separation and the condition (LB), and find other potential uses for the nonlinear operator extending the Gramian. These matters are the subject of ongoing work.

Figure 2: The open question in the source, page 35.

For the converse, the exponent $p - 1$ identifies the right intermediate space. The weighted Bergman space with density $(1 - |z|)^{p-3}$ has Carleson exponent $(p - 3) + 2 = p - 1$. A dyadic frequency calculation shows

$$\ell_A^p \hookrightarrow A_{p-3}^p.$$

On a block of N coefficients, the loss from ℓ^p coefficients to an $L^p(\mathbb{T})$ norm is $N^{1-2/p}$; its p -th power is N^{p-2} . Radial integration over the corresponding boundary layer contributes exactly $N^{-(p-2)}$. This cancellation is the core of the proof. A standard Whitney-box argument, included below, then embeds A_{p-3}^p into $L^p(\mu)$ precisely under the displayed box condition.

3 Main theorem

Theorem 1 (Complete characterization). *Let $2 < p < \infty$ and let μ be a finite positive Borel measure on $\mathbb{D}$. Then (1) holds for all $f \in \ell_A^p$ if and only if $[\mu]_{p-1} < \infty$. Quantitatively,*

$$c_p[\mu]_{p-1} \leq \sup_{0 \neq f \in \ell_A^p} \frac{\int_{\mathbb{D}} |f|^p d\mu}{\|f\|_{\ell_A^p}^p} \leq C_p[\mu]_{p-1},$$

where $c_p, C_p > 0$ depend only on p (and on the harmless convention used for Carleson squares).

Necessity

Assume (1) with best constant M . Fix a square $S(\theta, h)$ with h small. Choose an integer $N \asymp h^{-1}$ so that $Nh \leq 1/16$, and put

$$F_{N,\theta}(z) = N^{-1/p} \sum_{n=0}^{N-1} e^{-in\theta} z^n.$$

Its ℓ_A^p norm is one. If $z = re^{i(\theta+t)}$ belongs to $S(\theta, h)$, then $|t| \leq h/2$, $r \geq 1 - h$, and for $0 \leq n < N$,

$$r^n \geq (1 - h)^N \geq c, \quad |nt| \leq Nh/2 \leq 1/32.$$

Taking real parts after the rotation by $e^{-in\theta}$ gives

$$\left| \sum_{n=0}^{N-1} e^{-in\theta} z^n \right| \geq cN.$$

Consequently $|F_{N,\theta}(z)|^p \geq c_p N^{p-1}$ throughout the square, and

$$M \geq \int_{\mathbb{D}} |F_{N,\theta}|^p d\mu \geq c_p N^{p-1} \mu(S(\theta, h)).$$

Since $N \asymp h^{-1}$, this yields $\mu(S(\theta, h)) \leq C_p M h^{p-1}$. For h bounded away from zero, the same conclusion follows by testing with the constant function 1. Hence $[\mu]_{p-1} \leq C_p M$.

The critical coefficient-to-Bergman estimate

Let dA denote normalized area measure.

Lemma 2. *For $2 < p < \infty$ and every analytic polynomial $f(z) = \sum a_n z^n$,*

$$\int_{\mathbb{D}} |f(z)|^p (1 - |z|)^{p-3} dA(z) \leq C_p \sum_{n \geq 0} |a_n|^p. \quad (3)$$

Proof. Set $\beta = 1 - 2/p = (p - 2)/p > 0$. Apart from the constant coefficient, split f into dyadic blocks

$$f_k(z) = \sum_{2^k \leq n < 2^{k+1}} a_n z^n, \quad b_k = \left(\sum_{2^k \leq n < 2^{k+1}} |a_n|^p \right)^{1/p}.$$

For a trigonometric polynomial P with at most N frequencies, interpolation between $\|P\|_\infty \leq N^{1/2} \|P\|_2$ and the identity on L^2 gives

$$\|P\|_p \leq N^{1/2-1/p} \|P\|_2.$$

Applying this at radius r , and then using $\|c\|_2 \leq N^{1/2-1/p}\|c\|_p$, yields

$$M_p(r, f_k) \leq C_p 2^{k\beta} r^{2^k} b_k, \quad (4)$$

where $M_p(r, g) = (2\pi)^{-1/p} \|g(re^{it})\|_{L^p(0, 2\pi)}$.

For $r \geq 1/2$, let j be determined by $2^{-j-1} < 1-r \leq 2^{-j}$. Since $r^{2^k} \leq \exp(-c2^{k-j})$, Minkowski's inequality and (4) give

$$M_p(r, f) \leq |a_0| + C_p G_j, \quad G_j = \sum_{k \geq 0} 2^{k\beta} e^{-c2^{k-j}} b_k.$$

The radial weighted measure of this shell is comparable to $2^{-j(p-2)} = 2^{-j\beta p}$. Moreover,

$$2^{-j\beta} G_j \leq \sum_{k \leq j} 2^{-(j-k)\beta} b_k + \sum_{k > j} 2^{(k-j)\beta} e^{-c2^{k-j}} b_k.$$

Both right-hand kernels are summable as functions of $j-k$. Young's convolution inequality on $\ell^p(\mathbb{Z})$ therefore shows

$$\sum_{j \geq 1} 2^{-j(p-2)} G_j^p \leq C_p \sum_{k \geq 0} b_k^p.$$

This proves (3) on $1/2 \leq r < 1$. On $0 \leq r < 1/2$, Holder's inequality gives

$$|f(re^{it})| \leq \|(a_n)\|_{\ell^p} \left(\sum_{n \geq 0} r^{np'} \right)^{1/p'} \leq C_p \|(a_n)\|_{\ell^p},$$

so the remaining part follows as well. $\square$

The weighted Bergman Carleson step

Lemma 3. *Let $\alpha > -1$ and suppose a positive measure μ satisfies*

$$\mu(S(\theta, h)) \leq Kh^{\alpha+2} \quad (0 < h \leq 1).$$

Then every analytic f satisfies

$$\int_{\mathbb{D}} |f|^p d\mu \leq C_{p,\alpha} K \int_{\mathbb{D}} |f(z)|^p (1-|z|)^\alpha dA(z). \quad (5)$$

Proof. Decompose a neighborhood of $\partial\mathbb{D}$ into dyadic Whitney boxes Q of angular and radial side length h_Q , and let Q^* be a fixed enlargement. Each Q lies in a Carleson square of side comparable to h_Q , hence $\mu(Q) \leq CKh_Q^{\alpha+2}$. Since $|f|^p$ is subharmonic,

$$\sup_Q |f|^p \leq Ch_Q^{-2} \int_{Q^*} |f|^p dA \leq Ch_Q^{-\alpha-2} \int_{Q^*} |f(z)|^p (1-|z|)^\alpha dA(z).$$

The last comparison is valid also for $-1 < \alpha < 0$, because $1-|z| \asymp h_Q$ on Q^* . Thus

$$\int_Q |f|^p d\mu \leq CK \int_{Q^*} |f(z)|^p (1-|z|)^\alpha dA(z).$$

The enlarged boxes have bounded overlap. Summing, and treating a fixed central disk by the same submean estimate and the case $h = 1$, proves (5). $\square$

Completion of the proof

Assume $[\mu]_{p-1} < \infty$ and take $\alpha = p - 3 > -1$. Since $\alpha + 2 = p - 1$, Lemma 3 followed by Lemma 2 gives, for every analytic polynomial,

$$\int_{\mathbb{D}} |f|^p d\mu \leq C_p [\mu]_{p-1} \int_{\mathbb{D}} |f(z)|^p (1 - |z|)^{p-3} dA(z) \leq C_p [\mu]_{p-1} \|f\|_{\ell_A^p}^p.$$

For a general $f \in \ell_A^p$, its Taylor polynomials converge pointwise on $\mathbb{D}$ and in ℓ_A^p . Fatou's lemma extends the same estimate to f . Together with the necessity argument, this proves Theorem 1. $\square$

4 The exact answer to equation (7.2)

Corollary 4. *Let $2 < p < \infty$, $1/p + 1/q = 1$, and let (z_k) be distinct points of $\mathbb{D}$. The source inequality*

$$\sum_{k \geq 0} (1 - |z_k|^q)^{p-1} |f(z_k)|^p \leq K \|f\|_{\ell_A^p}^p$$

holds for every $f \in \ell_A^p$ if and only if

$$\sup_{\theta \in \mathbb{R}, 0 < h \leq 1} h^{-(p-1)} \sum_{z_k \in S(\theta, h)} (1 - |z_k|^q)^{p-1} < \infty. \quad (6)$$

Proof. If the source inequality holds, testing with the constant function shows that the atomic measure (2) is finite. Conversely, condition (6), together with a finite covering of the disk by Carleson squares of side one, also implies finiteness. The assertion now follows by applying Theorem 1 to that atomic measure. $\square$

Thus the open norm inequality is characterized by an explicit $(p-1)$ -Carleson packing condition. The ordinary Carleson condition found in the source is weaker, exactly as its counterexample indicates.

5 Verification and novelty check

Internal verification. The proof is scale-invariant at both decisive points. A normalized block of N equal coefficients has boundary L^p size $N^{1-2/p}$, whose p -th power is N^{p-2} ; the weight $(1-r)^{p-3} dr$ assigns mass $N^{-(p-2)}$ to the layer $1-r \asymp N^{-1}$. The diagnostic script `code/check_scaling.py` samples the Dirichlet lower bound and checks this exponent cancellation for $p = 2.2, 2.5, 3, 4, 6$ and $N = 8, 16, 32, 64, 128$. These computations are sanity checks, not proof.

Bounded literature search. On 22 July 2026 we searched the run indexes and web/arXiv results for 2210.05823, “ ℓ_A^p Carleson measure”, “analytic coefficient space ℓ^p Carleson”, “ $p-3$ weighted Bergman ℓ^p coefficients”, and the proposed exponent “ $|I|^{p-1}$ ”. The search found the source paper and general weighted Bergman Carleson theorems, but no paper stating the characterization in Theorem 1 or Corollary 4. This is a bounded, not exhaustive, novelty check.

Limitations and reviewer focus. The packet addresses the measure/norm-inequality question and therefore the source condition (UB) through Theorem 7.1 of the source. It does not settle the separate requested connection between weak separation and condition (LB), nor questions about other uses of the nonlinear Gramian. The main review point is Lemma 2, especially the negative but integrable weight $p-3 \in (-1, 0)$ when $2 < p < 3$.

Recommendation. Send for expert review as a full solution candidate to the explicit norm inequality (7.2). If accepted, the natural next step is to state the packing condition directly as a characterization of (UB) and investigate whether it interacts with the source’s weak-separation conditions.

References

- [1] R. Cheng and C. Felder, *On Interpolation by Functions in ℓ_A^p* , arXiv:2210.05823, 2022.
- [2] P. Duren and A. Schuster, *Bergman Spaces*, Mathematical Surveys and Monographs 100, American Mathematical Society, 2004.